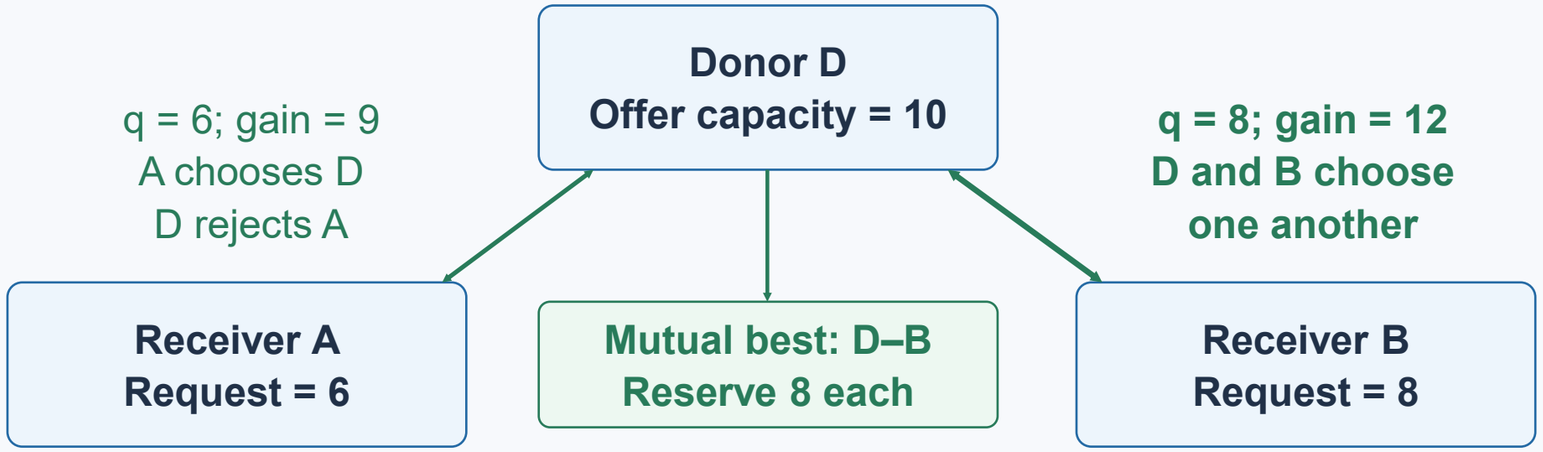
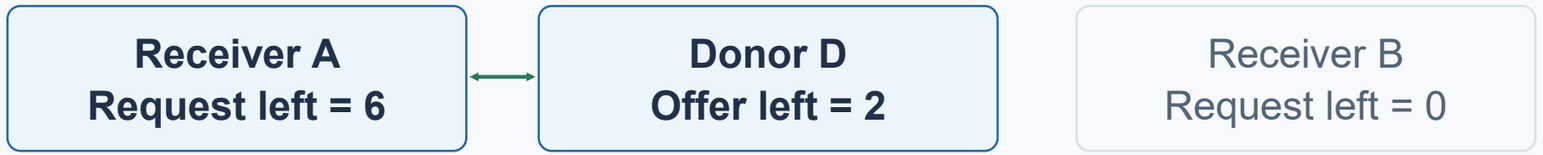
Peer matching: two receivers compete for one donor

Illustrative K = 0 example (simulator default K = 10): G(q) = 1.5q. All promises are pre-demand.

ROUND 1 | A and B both request from D



ROUND 2 | reservations leave D with 2 units



D–A: q = 2, gain = 3; mutual best → reserve 2. B has no residual request.

Fixed matching rules

- Common total edge ranking
- Choose positive-gain best edge
- Confirm only mutual choices
- At most one partner per round
- Reserve before the next round

Conditional promise ledger

D–B: 8 D–A: 2
Total reserved at D = 10

Promises are not dispatch.
No receive-forward.
Recheck after demand.